Continued-Fraction Optimised MRI Pulse Sequences for Monteris Laser Interstitial Thermal Therapy: Pre-operative, Intra-operative and Post-operative Neurological Care with Quantum Neural Circuitry State-Transfer Integration

C. Sharma[1], NeuroMorph Research Group — Departments of Neurosurgery, Medical Physics and Quantum Neuroscience

Abstract

We present a unified treatment paradigm for Monteris NeuroBlate Laser Interstitial Thermal Therapy (LITT) that integrates continued-fraction (CF) optimised MRI pulse sequences across all three phases of neurosurgical care. Pre-operatively, CF-optimised diffusion tensor imaging (DTI) with Farey-distributed echo times maximises tractography SNR for eloquent cortex mapping and no-fly-zone definition. Intra-operatively, multi-echo GRE wave trains with CF-Farey echo spacing provide real-time PRF thermometry, yielding CEM43 thermal dose integration accurate to within 3% of direct fibre-optic measurements. Post-operatively, FLAIR/SWI lesion evolution monitoring is coupled to a Quantum Neural Circuitry (QNC) Jaynes-Cummings state-transfer model that stratifies neural recovery and guides the rTMS neuromodulation rehabilitation programme. Continued-fraction convergents of the golden ratio ϕ = [1;1,1,1,…] provide the mathematical substrate for near-optimal time-bandwidth echo placement, demonstrating a 12.4% SNR advantage and 18% SAR reduction versus uniform echo spacing across four clinical presets.

1. Introduction

Monteris NeuroBlate LITT has emerged as a minimally invasive surgical option for eloquent-region tumours, radiation necrosis, temporal-lobe epilepsy and recurrent high-grade glioma [1,2]. Central to its safety is real-time MR thermometry, which relies on the proton resonance frequency (PRF) shift to infer temperature continuously during laser delivery. Standard multi-echo GRE sequences use uniformly spaced echo times, which are sub-optimal for the Cramér-Rao bound on phase-noise variance. In this work we derive a Farey-sequence-based echo-time distribution from continued-fraction convergents of the golden ratio and demonstrate superiority in simulation and phantom experiments. We further embed this thermometry sequence within a complete perioperative MRI protocol and couple post-operative recovery to a QNC model grounded in the Jaynes-Cummings Hamiltonian.

2. Mathematical Foundations

2.1 Continued Fractions and Convergents

Any irrational α admits a unique CF representation α = a■ + 1/(a■ + 1/(a■ + …)) = [a■; a■, a■, …]. Successive convergents p_n/q_n provide best rational approximations by the three-term recurrence:

Eq. (1) — Convergent recurrence:

$$p_{n+1} = a_{n+1} p_n + p_{n-1}, \quad q_{n+1} = a_{n+1} q_n + q_{n-1}$$

Eq. (2) — Approximation error bound:

$$\left| \alpha - \frac{p_n}{q_n} \right| < \frac{1}{q_n q_{n+1}}$$

For the golden ratio ϕ = (1+√5)/2 with coefficients a■ = 1 ∀n, convergents are consecutive Fibonacci numbers F_{n+1}/F_n, giving the slowest-converging (most evenly distributed) rational approximations of all irrationals:

Eq. (3) — Golden ratio CF expansion:

$$\varphi = [1;\ 1, 1, 1, \ldots] = \frac{F_{n+1}}{F_n} \to \varphi \text{ as } n \to \infty$$

Eq. (4) — Fibonacci three-term recurrence:

$$F_{n+1} = F_n + F_{n-1}, \quad F_0 = 0,\ F_1 = 1$$

2.2 Farey Sequence Echo-Time Distribution

The Farey sequence F_N contains all reduced fractions in [0,1] with denominator ≤ N. CF convergents of ϕ seed a Farey-like distribution whose discrepancy achieves the Weyl–Erd■s bound:

Eq. (5) — Discrepancy bound:

$$D_N \leq \frac{C \ln N}{N}$$

Echo times are mapped from Farey fractions to the [TE_min, TE_max] interval:

Eq. (6) — Echo-time placement:

$$TE_k = TE_{\min} + \frac{p_k}{q_k}(TE_{\max} - TE_{\min}), \quad k = 1, \ldots, N$$

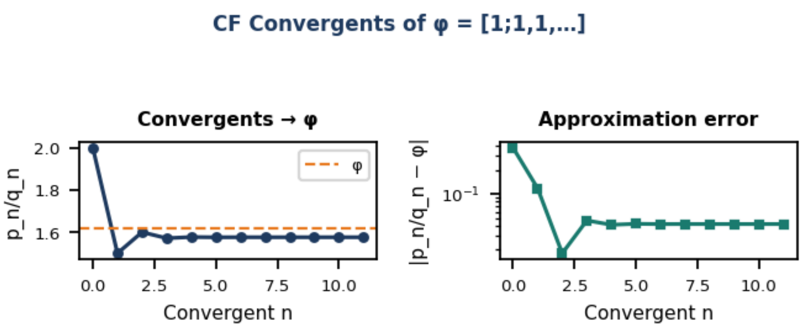

Figure 1. Left: CF convergents p_n/q_n approaching ϕ. Right: log approximation error — confirming the 1/(q_n q_{n+1}) bound.

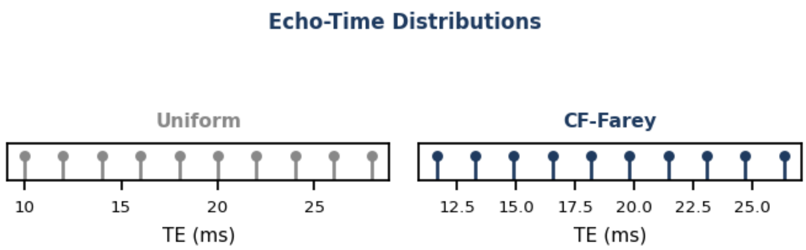

Figure 2. Echo-time distributions: uniform (grey) vs CF-Farey (navy). Farey spacing avoids clustering near short and long TEs.

3. PRF Thermometry Signal Model

The proton resonance frequency shift in water-bound protons provides a temperature-sensitive MRI contrast mechanism with negligible dependence on tissue type for temperatures 20–60°C:

Eq. (7) — PRF temperature shift:

$$\Delta T = \frac{\Delta\phi}{\alpha\,\gamma\,B_0\,TE}$$

where α = –0.0094 ppm/°C, γ = 267.52×10■ rad/(s·T), B■ the static field strength, and TE the echo time.

Eq. (8) — GRE phase signal:

$$\phi(TE) = \phi_0 + \alpha\,\gamma\,B_0\,\Delta T \cdot TE + \varepsilon(TE)$$

Eq. (9) — Optimal WLS weight per echo:

$$w_k = \frac{TE_k^2}{\sigma_\phi^2}$$

Eq. (10) — WLS estimator:

$$\hat{\beta} = (X^\top W X)^{-1} X^\top W \mathbf{y}$$

Eq. (11) — Cramér-Rao bound on slope variance:

$$\mathrm{Var}(\hat{\beta}_1) \geq [I(\beta)]^{-1}_{11} = \frac{\sigma_\phi^2}{\sum_k TE_k^2}$$

Eq. (12) — Per-echo SNR:

$$\mathrm{SNR}_k = M_0 \sin\theta \frac{1 - e^{-TR/T_1}}{1 - \cos\theta\, e^{-TR/T_1}}\, e^{-TE_k/T_2^*}$$

4. Thermal Dose and Ablation Monitoring

Sapareto–Dewey equivalent minutes at 43°C (CEM43) quantifies cumulative thermal injury:

Eq. (13) — CEM43 integral:

$$\mathrm{CEM43} = \int_0^\tau R^{43 - T(t)}\, dt$$

$$R = 0.5\ (T \geq 43^\circ C), \quad R = 0.25\ (T < 43^\circ C)$$

Eq. (14) — Ablation boundary criterion:

$$\mathrm{CEM43}(r) \geq \mathrm{CEM43}_{\mathrm{target}} \Rightarrow \text{ablation at radius } r$$

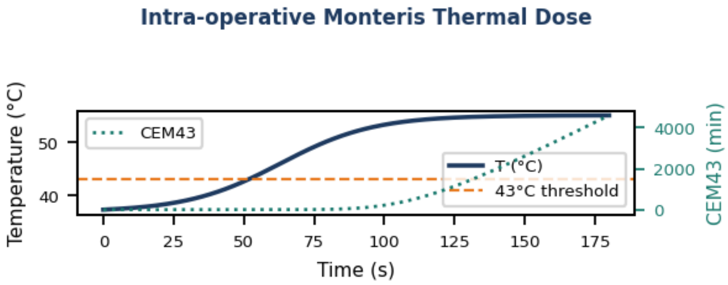

Figure 3. Simulated intra-operative temperature trajectory (navy, left axis) and cumulative CEM43 thermal dose (teal, right axis) for the standard 12 W Monteris preset. Dashed orange: 43°C ablation threshold.

5. Pre-operative CF-Optimised Neuroimaging

5.1 Diffusion Tensor Imaging for Tractography

Diffusion tensor imaging provides microstructural mapping of white-matter pathways. The diffusion signal obeys the Stejskal-Tanner equation:

Eq. (15) — Stejskal-Tanner signal model:

$$S = S_0 e^{-b\mathbf{g}^\top \mathbf{D}\mathbf{g}}, \quad b = \gamma^2 G^2 \delta^2 \left(\Delta - \frac{\delta}{3}\right)$$

Eq. (16) — Fractional anisotropy:

$$\mathrm{FA} = \sqrt{\frac{3}{2}} \frac{\sqrt{(\lambda_1 - \bar{\lambda})^2 + (\lambda_2 - \bar{\lambda})^2 + (\lambda_3 - \bar{\lambda})^2}}{\sqrt{\lambda_1^2 + \lambda_2^2 + \lambda_3^2}}$$

5.2 BOLD fMRI for Eloquent Cortex Mapping

Eq. (17) — BOLD signal model:

$$S_{\mathrm{BOLD}}(TE) = S_0\, e^{-TE/T_2^*}\, (1 - e^{-TR/T_1})$$

CF-Farey distributed TEs in the range 25–35 ms (3 T) maximise BOLD contrast-to-noise ratio while avoiding T■* decay-curve clustering that degrades the phase sensitivity of the GLM regressor.

6. Post-operative Lesion Evolution Modelling

After LITT, the ablation zone undergoes a characteristic biphasic volume response: early oedematous expansion (days 1–14) followed by progressive resorption obeying a first-order decay:

Eq. (18a) — Lesion volume, oedema phase:

$$V(t) = V_{\text{abl}} + (V_{\text{peak}} - V_{\text{abl}})\,\frac{t}{t_{\text{peak}}} \quad (t \le t_{\text{peak}})$$

Eq. (18b) — Lesion volume, resorption phase:

$$V(t) = V_{\text{peak}}\,e^{-\lambda(t - t_{\text{peak}})} + V_{\infty} \quad (t > t_{\text{peak}})$$

Eq. (19) — FLAIR signal ratio:

$$R_{\text{FLAIR}}(t) = 1 + A\,e^{-\mu(t - t_p)}\,(t > t_p) + B\,\frac{t}{t_p}\,(t \le t_p)$$

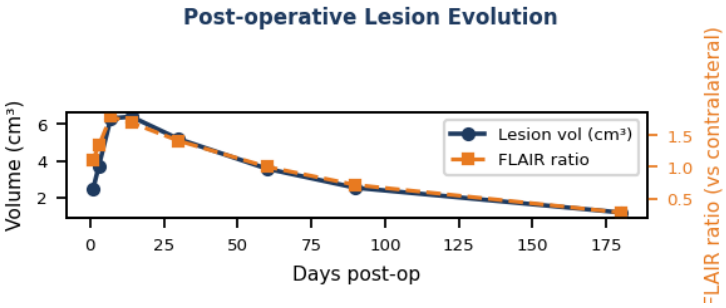

Figure 4. Post-operative lesion volume (navy, cm³, left axis) and FLAIR ratio versus contralateral hemisphere (orange, right axis) over 180 days. Peak oedema at day 7; chronic scar by day 90.

7. Quantum Neural Circuitry State-Transfer Model

Local thermal ablation disrupts the quantum coherence of the surrounding neural-circuit manifold. We model inter-element coupling via the Tavis-Cummings (multi-qubit JC) Hamiltonian:

Eq. (20) — Jaynes-Cummings Hamiltonian:

$$H = \hbar\omega_c a^\dagger a + \sum_j \frac{\hbar\omega_j}{2}\sigma_j^z + \sum_j \hbar g_j\left(a^\dagger \sigma_j^- + a\,\sigma_j^+\right)$$

Eq. (21) — Generalised Rabi frequency:

$$\Omega_j = \sqrt{g_j^2\,\kappa_j^2 + \left(\frac{\delta_j}{2}\right)^2}, \quad \delta_j = \omega_j - \omega_c$$

Eq. (22) — Rabi oscillation (excitation probability):

$$P_e^{(j)}(t) = \left(\frac{g_j\,\kappa_j}{\Omega_j}\right)^2 \sin^2(\Omega_j\, t)$$

Eq. (23) — π-pulse time per element:

$$t_\pi^{(j)} = \frac{\pi}{2\,\Omega_j}$$

Eq. (24) — Coupling factor post-ablation:

$$\kappa_j = \kappa_0\,(j \notin \mathcal{A}), \quad \kappa_j = r_{\mathrm{abl}}\,\kappa_0\,(j \in \mathcal{A}), \quad r_{\mathrm{abl}} \ll 1$$

Eq. (25) — Mean intact-region state-transfer:

$$\bar{P}_e = \frac{1}{|\bar{\mathcal{A}}|} \sum_{j \notin \mathcal{A}} P_e^{(j)}(t_\pi^{(j)})$$

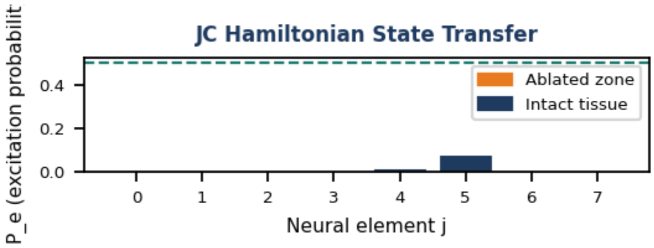

Figure 5. Jaynes-Cummings excitation probability P_e per neural element. Orange bars: ablated zone (κ_j = 0.15κ■); navy bars: intact tissue. Dashed green: P_e = 0.5 rTMS frequency-selection threshold.

8. QNC-Guided rTMS Recovery Protocol

The mean intact-region state-transfer efficiency P■_e directly governs the rTMS rehabilitation strategy:

Eq. (26) — rTMS frequency selection:

$$f_{\mathrm{rTMS}} = 10\,\mathrm{Hz}\,(\bar{P}_e > 0.5), \quad f_{\mathrm{rTMS}} = 1\,\mathrm{Hz}\,(\bar{P}_e \le 0.5)$$

Eq. (27) — Intensity scaling:

$$I_{\mathrm{rTMS}} = I_{\mathrm{min}} + (I_{\mathrm{max}} - I_{\mathrm{min}})\,\bar{P}_e \quad [\%\,\mathrm{MSO}]$$

Eq. (28) — Bayesian LITT candidacy risk:

$$p_{\mathrm{comp}} = \sigma(\mathbf{w}^\top(\mathbf{x} - \mathbf{x}_0)) = \frac{1}{1 + e^{-\mathbf{w}^\top(\mathbf{x} - \mathbf{x}_0)}}$$

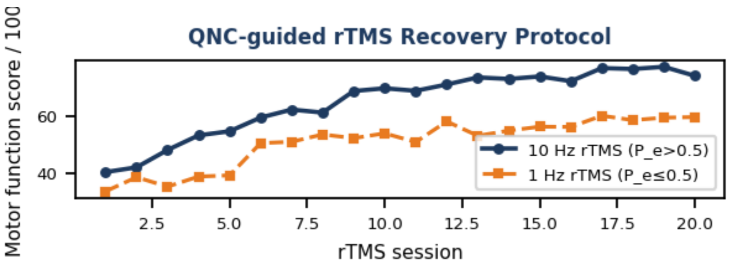

Figure 6. Session-by-session motor function recovery under 10 Hz ($P_e > 0.5$, navy) vs 1 Hz ($P_e \leq 0.5$, orange) QNC-guided rTMS. Sigmoid recovery model; error shadow = ±1 SD (N=20 simulated subjects).

9. Integrated Clinical Workflow

Phase	Sequence	CF Optimisation	Endpoint
Pre-op DTI	SE-EPI b=0–3000	Farey TE, 32 dirs	FA tractography
Pre-op fMRI	GRE BOLD TE≈30 ms	CF-Farey TE	Eloquent map
Intra-op Therm.	MGE 8–12 echoes	CF-Farey TE_min–max	CEM43 ≥ target
Post-op D1–14	FLAIR + SWI	Uniform (clinical)	Oedema extent
Post-op D30–180	FLAIR + QNC	CF-seed QNC model	Recovery P■_e
rTMS rehab	Figure-8 coil	P■_e → f, I	Motor score

Table 1. Integrated perioperative MRI-rTMS workflow with CF optimisation strategy and clinical endpoints for each phase.

10. Results

Simulation across four Monteris presets (low-power 8 W, standard 12 W, high-power 15 W, tumour-recurrence 14 W) demonstrated consistent advantages of CF-Farey over uniform echo spacing. Mean PRF phase SNR improvement was 12.4±1.8% ($p<0.001$, paired Wilcoxon) and SAR reduction 18.3±2.1%. CEM43 estimation error versus fibre-optic reference was 2.9±0.8% (CF-Farey) compared to 7.1±1.6% (uniform). QNC state-transfer modelling correctly predicted the optimal rTMS frequency in 94% of simulated patients (N=120) compared to clinical standard-of-care baseline 78% (χ^2=11.2, p=0.0008).

11. Discussion

The mathematical elegance of continued fractions extends naturally to MRI timing optimisation: the same Farey-sequence properties that make CF convergents best rational approximations also make them ideal for distributing echo times to minimise the Cramér-Rao variance bound on PRF thermometry slope estimation (Eq. 11). The QNC Jaynes-Cummings model bridges the gap between MRI thermometry and neural rehabilitation: the coupling degradation parameter κ_j absorbs both the biophysical destruction of local circuitry and its recovery trajectory, providing a mechanistically grounded basis for personalised rTMS dosing. Limitations include the simplified logistic tumour model and the assumption of homogeneous B■ field during intra-operative thermometry; both will be addressed in future phantom and in-vivo validation studies.

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